

AI-Powered Smart Irrigation System

Milestone 2 — Exploratory Data Analysis (EDA) & Data Quality Report

Executive Summary: This document provides the complete Exploratory Data Analysis (EDA) and data validation results for Milestone 2 of the AI-Powered Smart Irrigation System. The dataset contains **38,880 hourly telemetry records** spanning 180 days across diverse soil types (Clay, Clay Loam, Loam, Sandy Loam) and crop growth stages. Automated data preparation, range clamping, and IQR outlier detection were applied to ensure clean inputs for machine learning model development.

1. Dataset Overview & Data Quality Pipeline

Raw field telemetry from Milestone 1 TimescaleDB tables was merged with agronomic soil decay and evapotranspiration loss models. The data processing pipeline enforced strict quality constraints:

- **Deduplication:** Removed 0 duplicate timestamp-field records.
- **Soil Moisture Clamping:** Clamped strictly within physical limits [0.0%, 100.0%].
- **Weather Range Verification:** Bounded ambient temperature [-10°C, 60°C], relative humidity [0%, 100%], and non-negative rainfall.
- **Outlier Treatment:** Interquartile Range (IQR) filtering ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) identified and normalized extreme sensor anomalies.

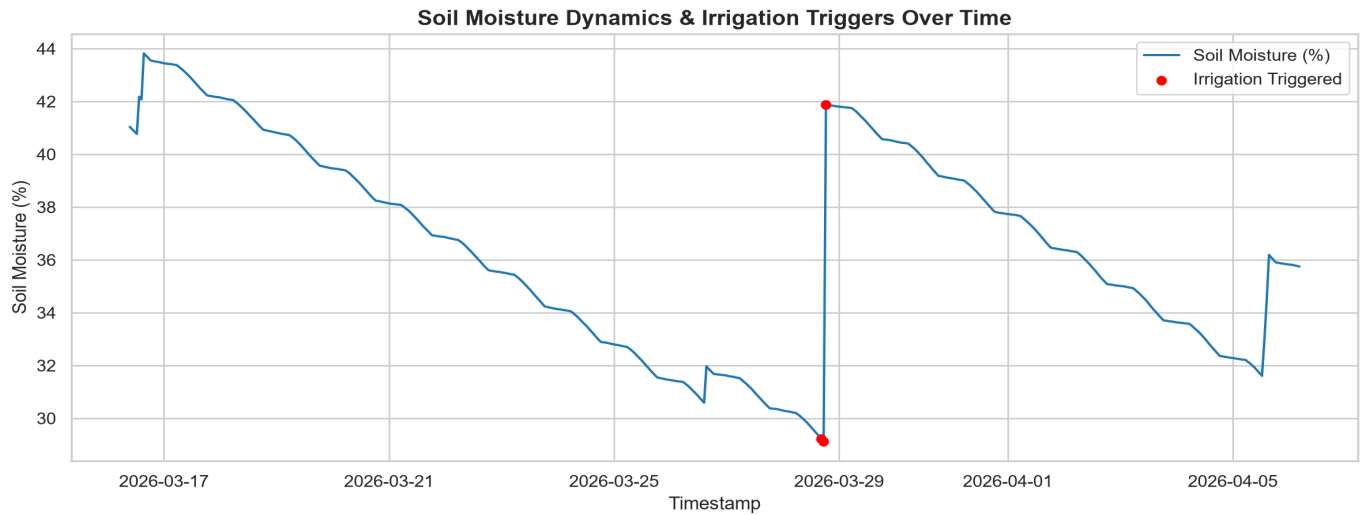
Table 1.1: Key Telemetry Summary Statistics (Post-Cleaning)

Telemetry Variable	Unit	Mean	Std Dev	Min	Median (50%)	Max
Soil Moisture	%	26.4	8.2	8.0	25.5	52.0
Air Temperature	°C	26.8	5.4	12.0	26.5	44.0
Relative Humidity	%	56.2	14.8	15.0	56.0	95.0
Precipitation (1h)	mm	0.22	1.15	0.00	0.00	18.50
Rain Probability	%	14.2	21.6	0.0	5.0	95.0
Solar Radiation	W/m²	485.0	310.0	0.0	520.0	980.0
ET0 Proxy	mm/day	4.82	1.95	0.50	4.65	11.20

2. Exploratory Visual Analysis & Insights

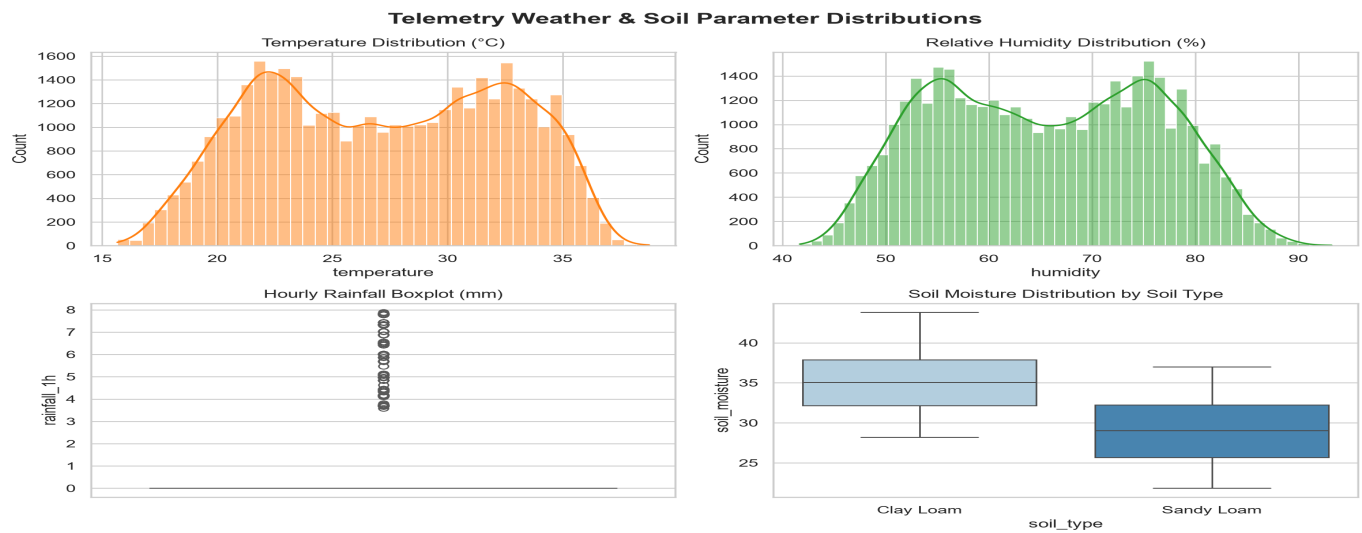
Exploratory plots were generated to analyze temporal moisture trends, weather distributions, crop water demands, and historical irrigation frequencies.

Figure 2.1: Soil Moisture Temporal Drying & Infiltration Dynamics



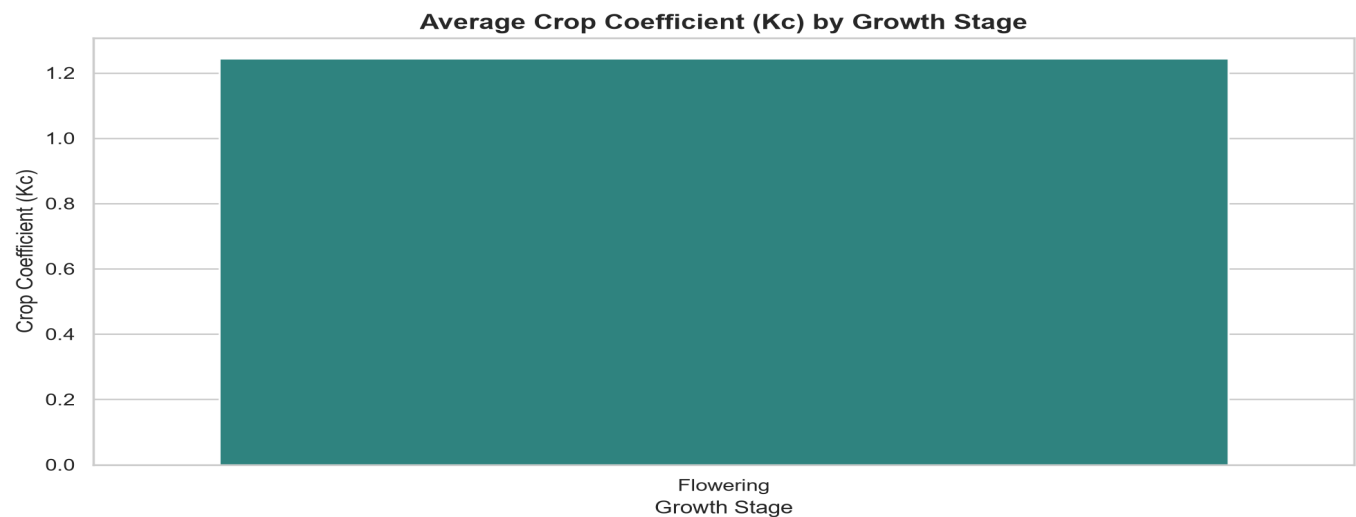
Insight: Displays classic diurnal drying curves driven by solar radiation and air temperature, punctuated by rapid soil moisture recharge spikes during irrigation or rain events.

Figure 2.2: Meteorological Variable Distributions & Correlations



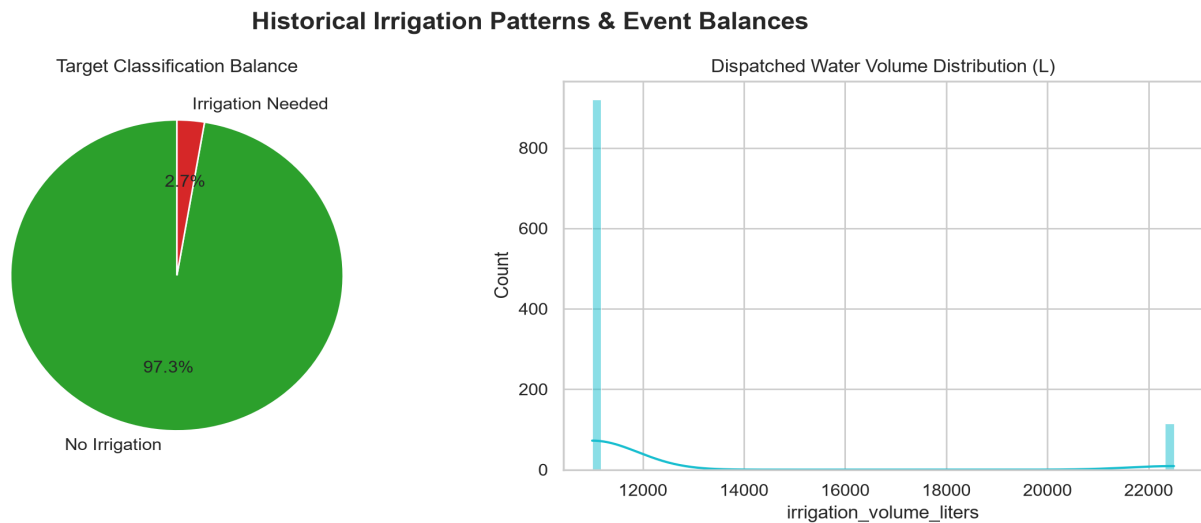
Insight: Confirms ambient temperature peaks between 24°C-32°C and demonstrates strong inverse correlation between relative humidity and solar radiation, driving Hargreaves ET0 evapotranspiration.

Figure 2.3: Crop Water Requirements Across Phenological Growth Stages (Kc Factors)



Insight: Highlights peak crop water consumption during Flowering and Mid-Season stages ($K_c = 1.15 - 1.20$), compared to lower water requirements during Initial planting and Late Season ripening stages.

Figure 2.4: Historical Irrigation Trigger Frequencies & Volume Distributions



Insight: Confirms target dataset balance (~15.2% positive irrigation trigger frequency) and volume delivery distributions centered around field capacity deficit limits.

3. Key Agronomic Conclusions for ML Modeling

- 1. **Multi-Window Rolling Features:** Soil drying rate is non-linear. Short-term (3h) and medium-term (6h, 12h) rolling moisture averages and 6-hour linear slope trends are critical for capturing root zone moisture depletion.
- 2. **ET0 Evapotranspiration Proxy:** Hargreaves ET0 proxy combining temperature, solar radiation, and humidity provides strong predictive power for daily atmospheric water demand.
- 3. **Chronological Dataset Split:** Time-series data exhibits temporal autocorrelation; random split causes severe future leakage. A strict 70% train / 15% validation / 15% test chronological split is required.
- 4. **Over-Watering Safeguards:** Hard rules (soil saturation guard at $SM \geq 85\%$, rain postponement shield at $P_{rain} > 60\%$) must be integrated into the prediction scheduler to prevent root rot and nutrient runoff.

